

NovaCart Sales Analytics – Key Insights & Business Recommendations

Project Overview

The NovaCart Sales Analytics project analyzes e-commerce sales, profitability, order behavior, product performance, regional performance, payment preferences, and customer performance using Power BI.

The dashboard was designed across three analytical views:

- Executive Business Performance
- Product & Order Performance
- Customer Performance

Recognized sales were used as the primary revenue metric so that cancelled and returned orders do not inflate reported sales.

Key KPIs

- **Recognized Sales:** ~₹690M
 - **Total Profit:** ~₹190M
 - **Profit Margin:** 27.48%
 - **Total Orders:** ~30K
 - **Total Customers:** ~5K
 - **Total Products:** 22
 - **Average Order Quantity:** 2.00 units
 - **Average Order Value:** ~₹25.57K
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Key Business Insights

1. NovaCart generated strong recognized revenue with healthy profitability

NovaCart generated approximately **₹690M in recognized sales** and **₹190M in profit**, resulting in an overall **profit margin of 27.48%**.

This indicates that the business maintained substantial profitability on successfully recognized sales.

2. Electronics is the dominant revenue-generating category

Electronics accounts for approximately **79% of recognized sales**, significantly outperforming the other product categories.

While this demonstrates strong demand for Electronics, it also indicates **high category concentration**, making overall business performance relatively dependent on this category.

3. A significant business slowdown occurred in the second half of the year

Sales remained comparatively stable between **January and June**, followed by a sharp decline beginning in **July**.

Sales recovered moderately afterward but remained substantially below first-half levels through the remainder of the year.

The dashboard establishes the decline but does not establish its underlying cause; additional analysis would be required to determine whether product demand, seasonality, customer activity, or another factor drove the change.

4. Lower order volume appears to be an important contributor to the sales decline

Monthly order volume was approximately **3K–3.5K orders during much of the first half**, falling to roughly **1.6K–1.8K per month during the second half**.

Because both sales and order volume decline around the same period, reduced transaction volume appears to be an important contributor to the lower second-half revenue.

5. Revenue is heavily concentrated among a small group of products

Product-level analysis shows that a relatively small number of products generate a substantial share of sales and profit.

Products such as the leading laptop and smartphone models appear among the strongest profit contributors, demonstrating the importance of NovaCart's high-value electronics portfolio.

6. Customer baskets typically contain around two units

The **Average Order Quantity is 2.00 units**, and the quantity distribution shows that single-unit and two-unit orders represent a substantial portion of transactions.

This provides an opportunity to increase basket size through relevant product bundles, cross-selling, and quantity-based promotions.

7. Completed orders dominate, but cancellations and returns still represent an improvement opportunity

The majority of NovaCart's approximately **30K orders are completed**, while the dashboard also identifies roughly **2K cancelled orders and 1K returned orders**.

Although cancelled and returned transactions have been excluded from recognized revenue, reducing these unsuccessful orders could improve realized sales and operational efficiency.

8. Geographic sales performance is relatively diversified

Unlike category performance, where Electronics dominates, sales are more evenly distributed across NovaCart's regions.

No single region appears overwhelmingly responsible for total revenue, reducing geographic concentration and providing opportunities for region-specific growth strategies.

9. Payment behavior is diversified

Customer sales are distributed across multiple payment methods, with **UPI representing the largest individual share at approximately 31%** in the displayed analysis.

The remaining sales are distributed across credit cards, debit cards, net banking, and cash-on-delivery, indicating that customers use a diverse mix of payment options.

10. High-sales customers and high-profit customers are not necessarily the same

Customer-level analysis separately identifies **Top Customers by Sales** and **Top Customers by Profit**.

This distinction is important because customers generating high revenue do not automatically generate the highest profit. Customer prioritization should therefore consider profitability alongside revenue.

Business Recommendations

1. Investigate the second-half decline

Perform a deeper drill-down of the July–December period across:

Product → Category → Region → Customer → Order Status

to determine which dimensions contributed most to the reduction in order volume and recognized sales.

The cause should be validated through further analysis rather than assumed from the trend alone.

2. Reduce dependence on Electronics while protecting its performance

Electronics is currently NovaCart's primary revenue driver, so the company should maintain appropriate inventory availability and promotional support for its strongest Electronics products.

At the same time, targeted campaigns and assortment strategies should be tested for lower-contributing categories to gradually diversify the revenue base.

3. Focus inventory on high-profit products

Products generating consistently high profit should receive greater attention in inventory planning and product availability.

Lower-performing products should be evaluated based on their sales, profitability, demand, and strategic role before additional inventory or promotional spending is committed.

4. Increase average basket size

With customers purchasing approximately **2 units per order on average**, NovaCart can test:

- Product bundles
- Cross-selling recommendations
- Quantity-based offers
- Complementary product suggestions

The objective should be to increase order value without materially reducing margins.

5. Analyze and reduce cancellations and returns

Cancelled and returned orders should be analyzed by product, category, region, payment method, and customer characteristics.

Identifying recurring patterns could help NovaCart improve order fulfillment and increase the proportion of placed orders that ultimately become recognized revenue.

6. Develop region-specific category strategies

Because overall regional sales are comparatively balanced, NovaCart should analyze **Category × Region** performance instead of applying a uniform strategy across all markets.

This can identify categories with unusually strong demand or profitability within individual regions.

7. Segment customers using both sales and profitability

Customer retention and loyalty initiatives should not be based solely on total sales.

NovaCart should identify customers who combine **high sales, high profit, and repeat purchasing behavior**, and prioritize these customers for retention campaigns and personalized offers.

8. Continue using recognized sales for management reporting

Cancelled and returned order values should remain separate from recognized revenue.

This ensures that headline KPIs such as **Total Sales, AOV, and Profit Margin** represent realized business performance rather than gross order value.

Short Conclusion

The NovaCart analysis shows a profitable e-commerce business generating approximately **₹690M in recognized sales at a 27.48% profit margin**. However, the analysis also highlights three important areas requiring management attention: **heavy dependence on Electronics, a substantial decline in second-half order activity, and opportunities to improve order completion and customer value**.

The recommended strategy is to protect the company's strongest products while investigating the second-half decline, reducing unsuccessful orders, diversifying category performance, and using customer profitability to guide retention and marketing decisions.

